

AGRITECH · HACKATHON 2026

Mandi-to-Market Supply Chain Optimizer

MandiPulse — a market-intelligence dashboard that finds where the mandi network is losing value for farmers, and why.

62,057

raw records ingested

5

source datasets fused

57

mandis · 3 states

₹63,195 Cr

market value mapped

Built with Python · pandas · Streamlit · Plotly | Live filters, 6 KPI cards, composite risk score & an offline AI analyst agent

A farmer's price is decided by things no farmer can see



Price opacity

The same crop clears at very different prices across mandis on the same day. Farmers cannot compare, so they accept whatever the nearest mandi offers.



MSP is a floor on paper

A Minimum Support Price exists for every major crop, but nothing in the system flags when sales actually clear below it.



Blind logistics

Produce moves to warehouses on trips nobody measures. Delay means spoilage, and spoilage is priced into the farmer's rate.

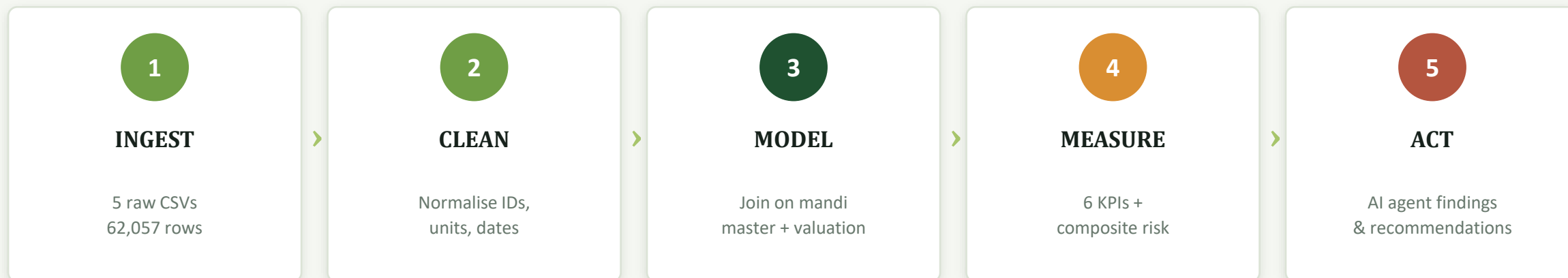


Unusable data

The records that could answer all of this exist — in five files with clashing units, IDs and date formats. Nobody can join them.

MandiPulse: one question, answered end to end

Guiding question: *“Where is the mandi network losing value for farmers, and why?”*



What ships today

7 tabs

Overview & Risk, AI Agent, Price & MSP, Arrivals, Logistics, Weather, Data Quality

9 filters

Date, state, district, mandi type, crop, mandi + 3 risk-weight sliders

1 agent

10 analyst checks and natural-language Q&A, running fully offline

Five datasets, one network



clean_mandi_master.csv

57 rows

The canonical hub — mandi_id, name, district, state, type, area. Every other file joins through it.



clean_price_and_msp.csv

12,000 rows

Daily min / max / modal price per mandi-crop, plus the MSP for that crop. Powers the MSP-compliance story.



clean_mandi_arrivals.csv

25,000 rows

Arrival lots: quantity, unit, crop, variety and farmer_count. Our volume and footfall base.



clean_transport_logistics.csv

10,000 rows

Mandi → warehouse trips: departure, arrival, transit hours, distance, vehicle, driver.



clean_weather_sensors.csv

15,000 rows

51 sensors logging temperature, rainfall and humidity — the spoilage-risk layer.

57

MANDIS

3

STATES

18

DISTRICTS

6

CROPS

Network make-up

Punjab	20 mandis
Haryana	20 mandis
Uttar Pradesh	13 mandis
APMC / Private / Direct	20 / 17 / 9
Window covered	Jan – Dec 2026

Every single file was entered by hand — and it shows

14

SPELLINGS OF ONE UNIT

Arrival quantity arrived as Qtl, qtl, Q, quintal, Quintals, KG, kg, KGS, Kilo, Kgs, MT, T, tonnes, Tonnes.

5

MANDI ID FORMATS

MANDI_001, MANDI-042, mandi_019, MANDI050 and a bare '038' — no join was possible before this was fixed.

6,881

SUB-₹50 PRICE ENTRIES

Decimal slips across min, max, modal and MSP. Real modal prices for these crops run ₹1,000–8,000/ctl.

4,048

CORRUPT TRANSIT HOURS

A 4-digit year leaked into the duration field, so trips showed '2026 hours' in transit.

7,593

UNPARSEABLE TIMESTAMPS

Weather rows mixed 28/08/2026, ISO datetimes and '...07:08:18 IST' in one column.

1,560

MISSING VEHICLE NUMBERS

Logistics rows with no vehicle recorded — kept, but flagged UNKNOWN rather than silently imputed.

Our rule: a row we cannot trust is excluded, never guessed at — and every exclusion is logged and shown in the dashboard's Data Quality tab.

Six deterministic, documented transforms

ID

Normalise every mandi ID

A regex pulls the first digit group and rebuilds it as MANDI_ONN, so MANDI-042, mandi_019, MANDI050 and '038' all collapse to one key. This single function is what makes the five-way join possible.

1:1

Reconcile units to one base

Quantities converted to quintals, distance to km, temperature to °C, rainfall to mm — regardless of how the unit was spelled in the source row.

<50

Reject impossible prices

Any price or MSP under ₹50/qtl is treated as a decimal-entry error and set to missing, so it can never pull an average down.

>200

Cap corrupt durations

Transit hours above 200 are year-leakage, not journeys. Dropped rather than clipped, because a fake value is worse than a gap.

||

Correct sign errors

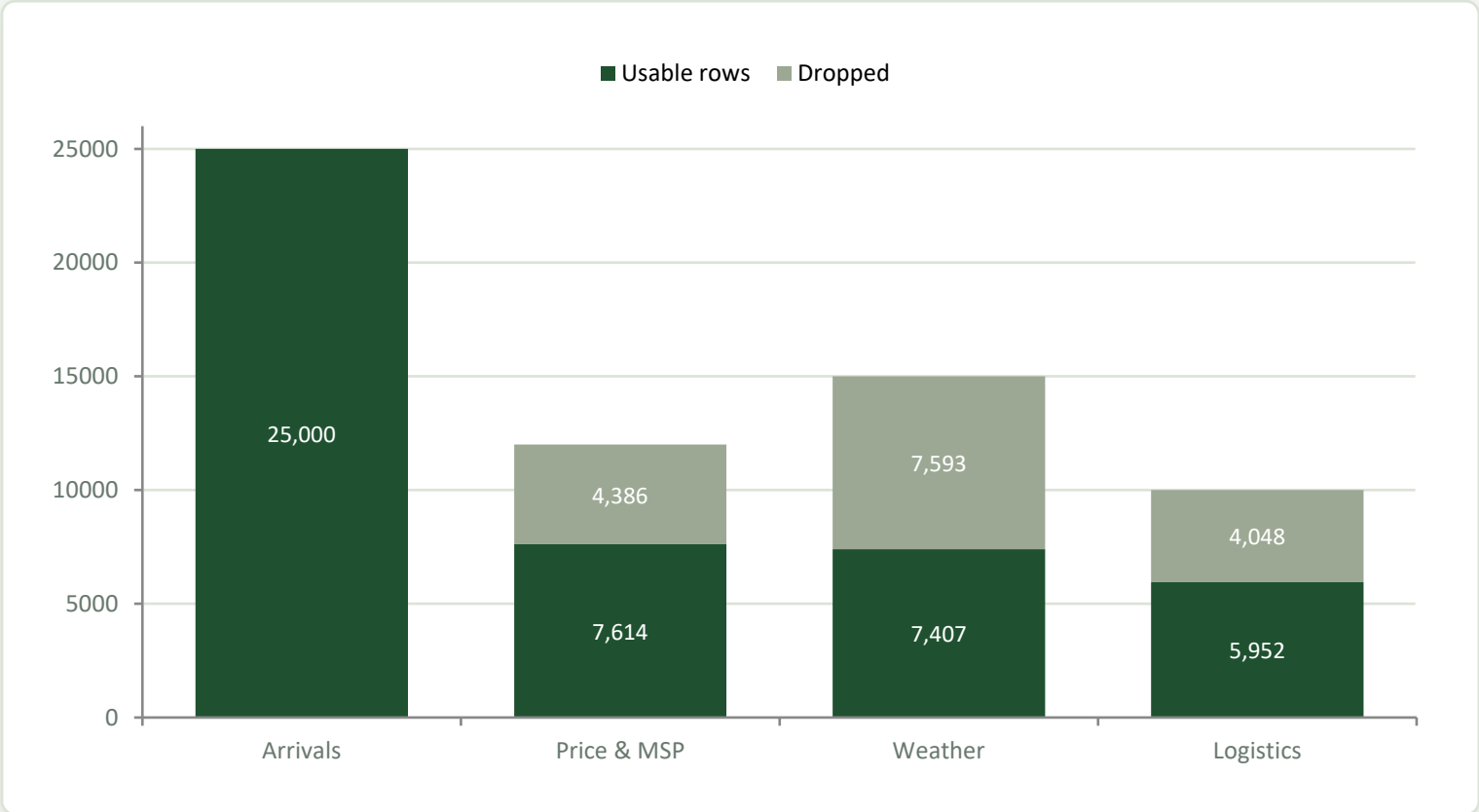
Negative arrival quantities are taken as data-entry sign flips and corrected with abs().

GEO

Single source of geography

Each file's own loose district column is discarded in favour of the mandi master's, so 'ludhiana' and 'Ludhiana' can never split one district in two.

46,030 of 62,057 rows survived — and we say so



46,030

rows usable after cleaning

25.8%

overall drop rate

0.0%

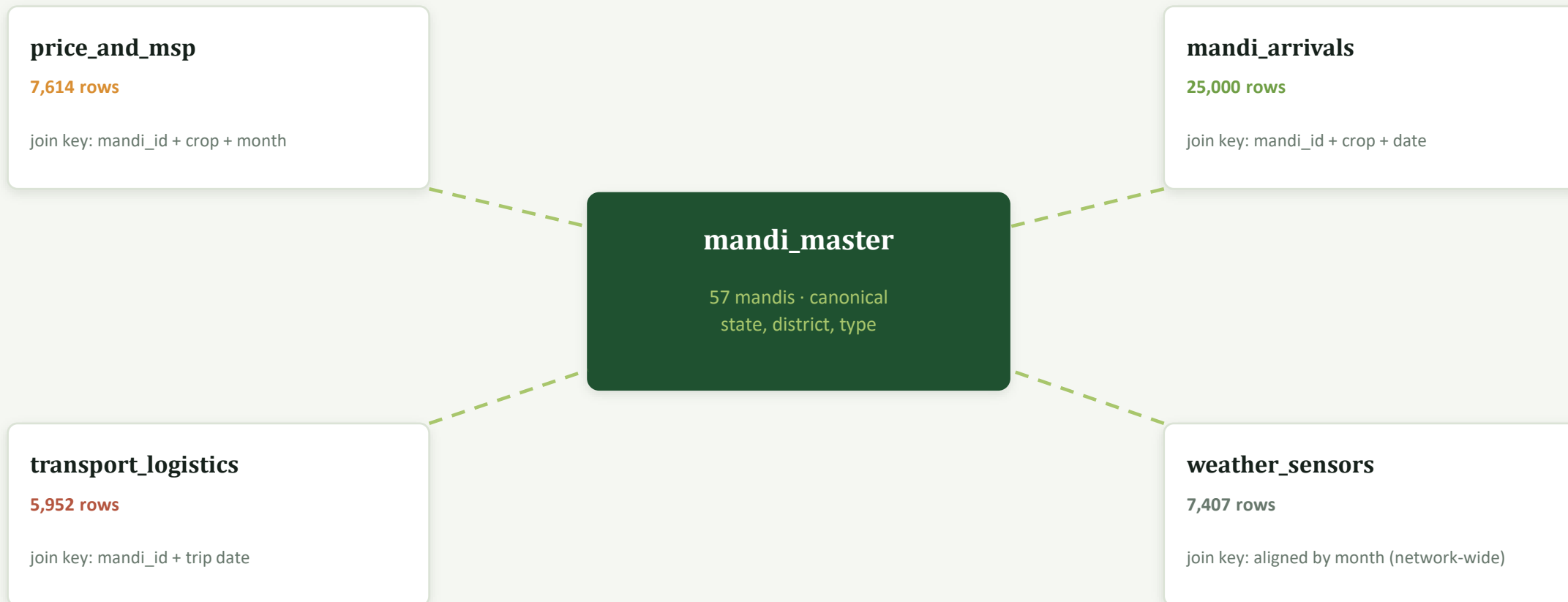
arrivals lost — the cleanest file

50.6%

weather lost to bad timestamps

Why this matters: the Data Quality tab exposes this table live, so a judge — or a mandi officer — can see the denominator behind every KPI instead of trusting a number.

mandi_id is the spine of the whole system



The derived table: arrivals × mandi-crop-month average modal price → **market_value**. Where a mandi has no price that month, it falls back to the crop's network-wide monthly average — so volume is never silently dropped from revenue.

Four metrics that did not exist in the raw data



market_value

```
arrival_quantity_qtl × avg modal price (mandi,  
crop, month)
```

Turns volume into rupees so 'value lost' becomes measurable. Labelled an estimate in the UI, not a ledger figure.



below_msp

```
modal_price < msp → farmer 'churn' risk
```

Converts a policy floor into a testable flag on every priced transaction. Aggregated, it becomes our headline KPI.



price volatility (CV)

```
std(modal_price) ÷ mean(modal_price) per mandi
```

A mandi can average a fair price and still be unpredictable. CV captures the risk a stable average hides.



weather_risk_index

```
mean(share of days >35°C, share in top-15%  
rainfall) × 100
```

One 0–100 spoilage-pressure number from two unrelated sensor streams.

Every one of these is recomputed against the current filter selection — nothing is precomputed or cached against a fixed slice.

Six cards that frame the entire story



Figures shown are the unfiltered network baseline. Every card recomputes the moment a filter changes.

A number without a direction is just trivia

Period-over-period logic

- Whatever window the date filter shows, we build the equal-length window immediately before it.
- Each KPI is recomputed on that prior window and shown as a % delta on the card.
- Change the filter and the comparison window moves with it — no fixed baselines.

Good direction is per-KPI

Revenue, arrivals, MSP compliance

up is green

Churn risk, transit time

down is green

Insufficient prior data

renders n/a, never 0%

Two accuracy bugs we found in testing — and fixed

The partial-month trap

Our data ends 8 December, so the trailing stub month looked like a 92% collapse in volume and flipped every reported trend. Fix: trend calculations now drop the trailing incomplete month.

2026-08: ₹7,473 Cr → 2026-12 stub: ₹627 Cr

The 'rice inside price' bug

Crop matching used a plain substring, and "rice" sits inside "p-rice". Asking the agent for the average price of wheat returned Rice. Fix: word-boundary regex matching.

"average price of wheat" → answered about Rice

Nine controls, one shared state, zero stale selections



Date range

Bounded to the real data extent; drives every KPI and the prior-period comparison.



State

3 options. Cascades into the district and mandi pools below it.



District

18 options, repopulated from the selected states.



Mandi type

APMC / Private / Direct — lets a judge compare governance models directly.



Crop

6 crops. Applied to price and arrivals, skipped on tables without a crop column.



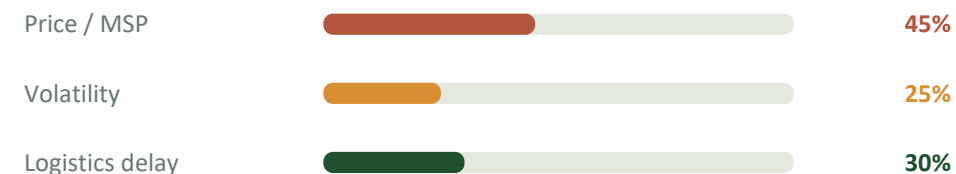
Mandi

57 options, narrowed by state + district + type before it is even rendered.

The “ALL” pattern

- Every dropdown carries a one-click ALL entry and defaults to it, so the app opens on the full network.
- Selections invalidated by a parent filter are silently dropped instead of throwing — pick a district, change the state, nothing breaks.

Risk-weight sliders



Plus a one-click “Reset all filters” that clears session state and reruns — essential when demoing live.

The composite Mandi Risk Score

$$\text{score} = 0.45 \cdot \text{norm}(\text{MSP breach rate}) + 0.25 \cdot \text{norm}(\text{price CV}) + 0.30 \cdot \text{norm}(\text{avg transit hrs}) \times 100$$

45%**Price risk**

Share of a mandi's priced sales that cleared below MSP. The farmer-facing harm, so it carries the most weight.

25%**Volatility**

Coefficient of variation of modal price. Unpredictability is its own cost — a farmer cannot plan around it.

30%**Logistics**

Mean transit hours to warehouse. Longer journeys mean more spoilage, which is priced back to the farmer.

**Min-max normalised**

Each component is scaled 0–1 within the current selection, so a percentage, a ratio and an hour count can be summed fairly.

**Median-filled, not zero-filled**

A mandi missing one component inherits the selection's median for it, so a data gap never masquerades as safety.

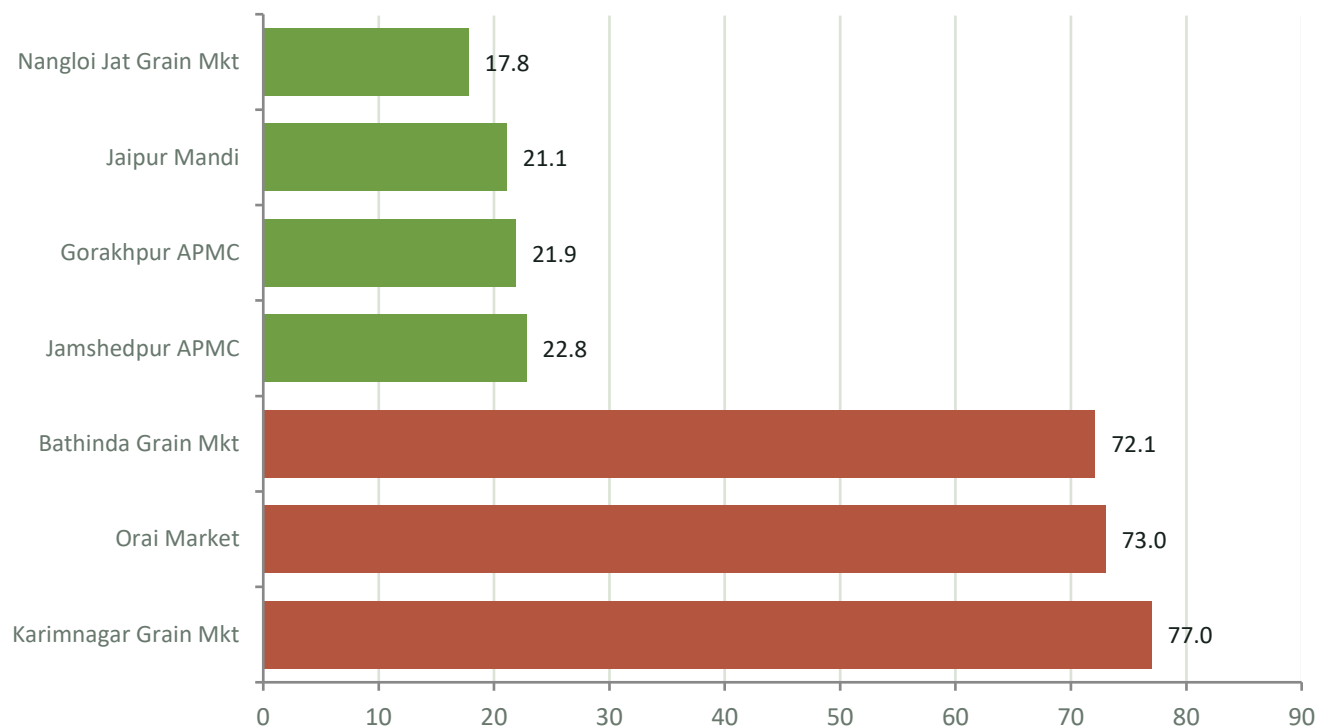
**User-tunable**

The three weights are live sliders. Judges can argue with our priorities and watch the leaderboard reorder.

**Banded 0–100**

≥66 high risk · 33–65 medium · <33 low, so the score reads as an action threshold rather than a raw number.

The network is not uniformly broken — 3 mandis are



Riskiest three vs. safest four, default weights (45 / 25 / 30)



Karimnagar Grain Market

Moga district, Punjab

77 / 100 — high risk

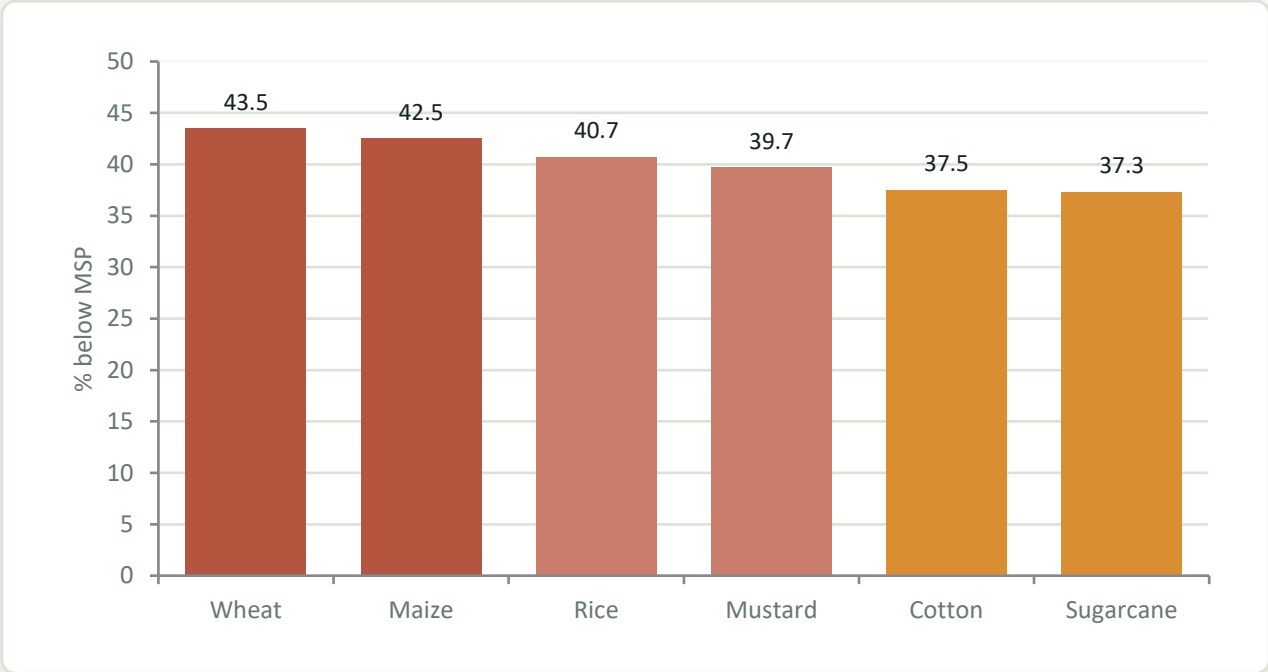
Ranks poorly on all three components at once, not just one.

Why it matters

- A 4.3× spread between worst and best mandi means intervention can be targeted, not universal.
- Fixing 3 mandis is a budget line. Fixing 57 is a policy debate.

Drill-down in the app: pick any mandi and a radar chart plots its price risk, volatility and transit time against the network average — so a mandi officer sees *which* lever to pull, not just that they scored badly.

The floor holds on average — and fails 2 times in 5



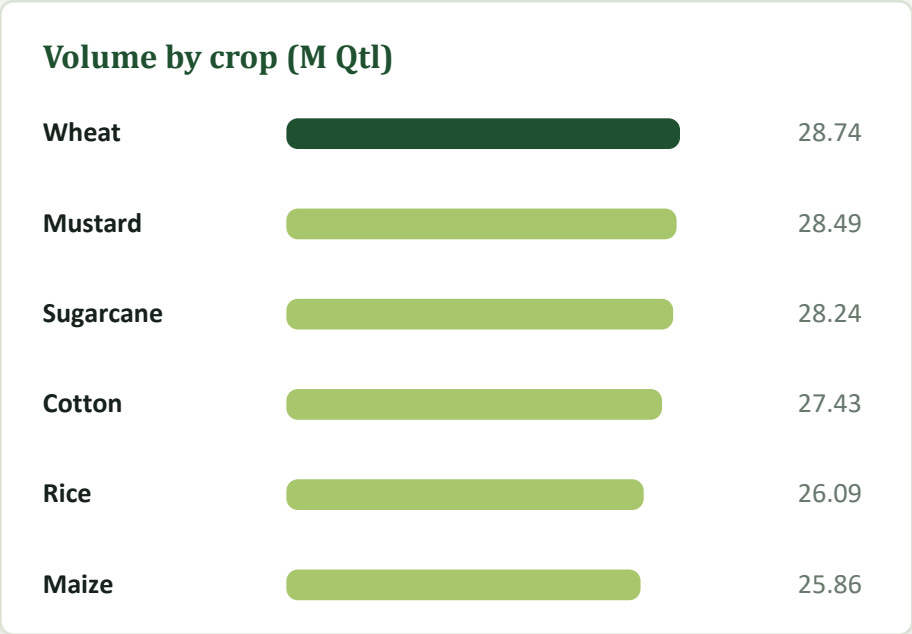
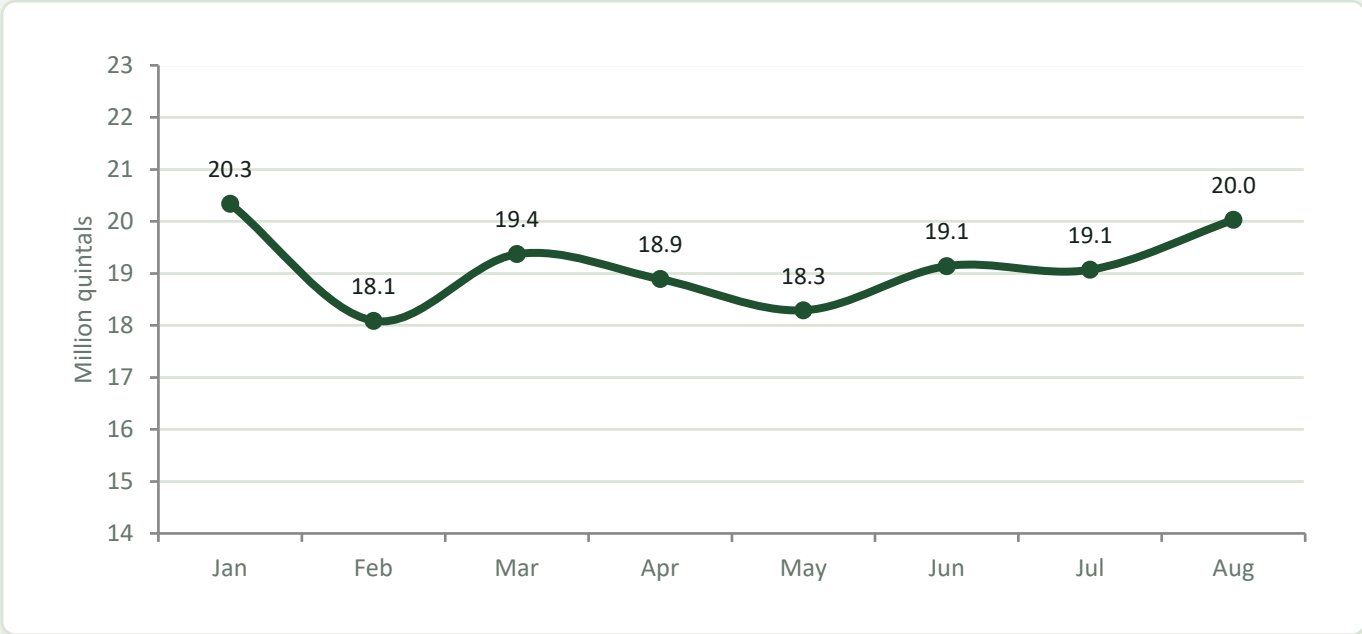
Average modal price vs MSP (₹/qtl)			
CROP	MODAL	MSP	GAP
Cotton	6,787	6,620	+167
Mustard	5,781	5,650	+131
Sugarcane	3,582	3,500	+82
Rice	2,232	2,183	+49
Maize	2,131	2,090	+41
Wheat	2,317	2,275	+42

MSP breach rate by crop • 4.488 priced rows carried an MSP

The insight the averages hide

Every crop's average modal price sits above its MSP — a policy dashboard would call that success. But 40.2% of individual transactions cleared below the floor. The mean is healthy because the breaches are offset by strong days; the farmer who sold on a breach day is not compensated by someone else's good day. Wheat is worst at 43.5%, and it is also our highest-volume crop — so the most exposed crop is the one most farmers depend on.

Volume is flat, broad and surprisingly decentralised



Complete months only — Sep–Dec are partial in the source and are excluded from trend logic

10.0%

of all volume sits in the top 5 mandis

Supply is genuinely decentralised — so price-discovery tooling has to reach the long tail, not just the big yards.

16.5 lakh

farmer visits recorded

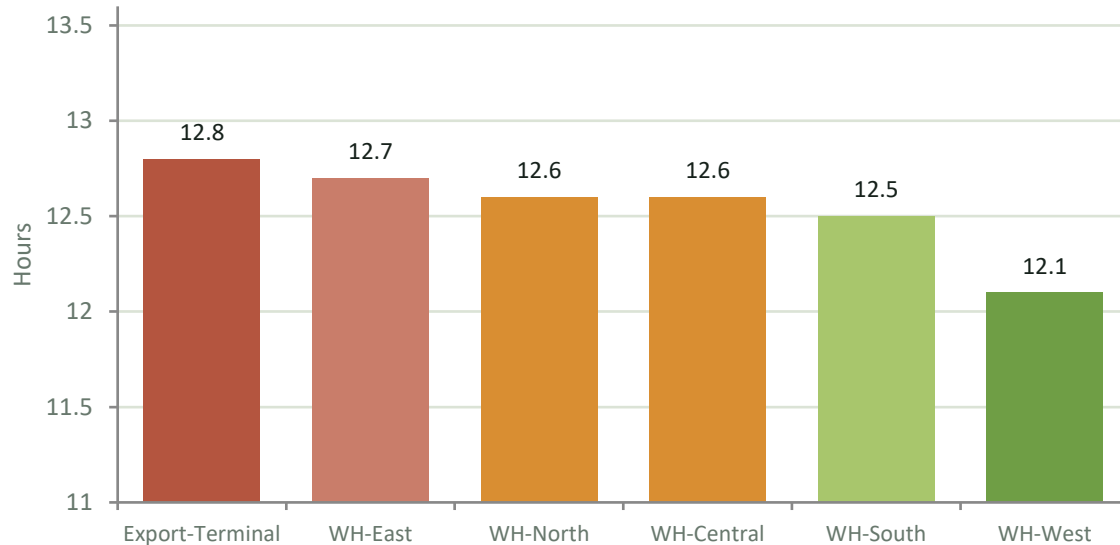
Arrivals carry farmer_count, so we can separate mandis serving many smallholders from those serving a few large sellers.

~11%

spread between top and bottom crop

No single crop dominates, which means an MSP fix must work across all six rather than being wheat-specific.

Distance explains almost nothing about delay



5,952 usable trips across 6 destinations

12.6 hrs

network average transit

611 km

average trip distance

$r = 0.025$

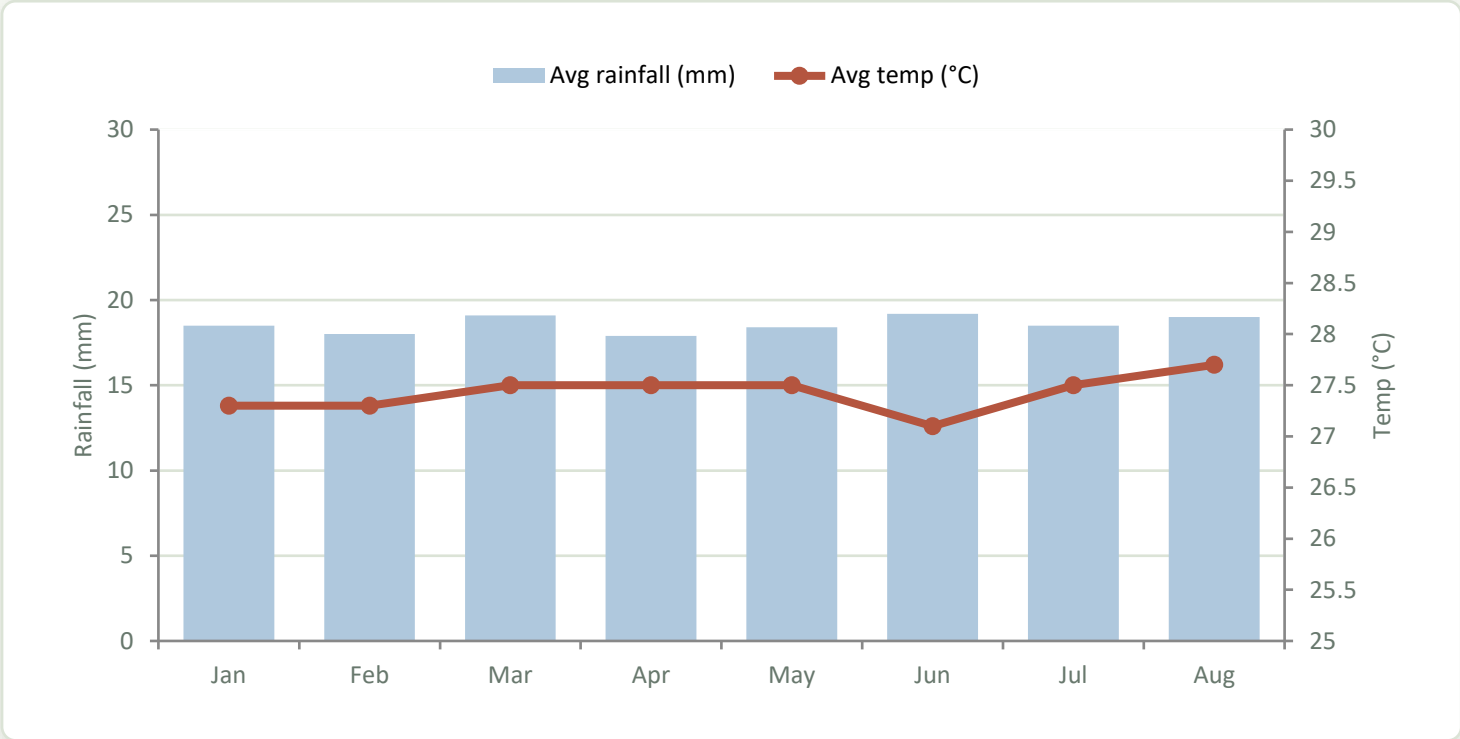
distance $\leftrightarrow$ transit correlation

Warehouses differ by only 0.7 hrs — there is no single bad depot to blame.

The counter-intuitive result

We expected transit time to scale with distance. It does not — the correlation is 0.025, effectively zero. Trips of 200 km and 1,000 km take about the same 12.6 hours. That points the finger away from routing and towards fixed overheads: loading queues, paperwork and dispatch waits at the mandi gate. It reframes the intervention from “find shorter routes” to “cut the hours a truck spends standing still”, and the dashboard’s distance-vs-transit scatter with its fitted trendline is where a judge can verify this in one look.

The spoilage layer: quiet now, but instrumented



15 / 100
weather risk index

16.4%
of readings above 35°C

27.4°C
mean temperature

18.6 mm
mean rainfall

An honest limitation

Sensors are not tagged to individual mandis in the source data, so this layer is network-wide and aligned by month. The dashboard says so on the tab rather than implying mandi-level precision we do not have.

51 sensors • 7,407 usable readings • complete months only

Conditions were stable across the window — a 0.6°C spread and no monsoon spike — so weather is not what drove this year's MSP breaches. That is a finding, not a gap: it rules out the easy external excuse and keeps the blame on price discovery and gate overheads. The rainfall-vs-arrivals scatter is already wired, so the moment a real shock year is loaded the correlation surfaces immediately.

An analyst agent that runs with no API key

10

PROACTIVE CHECKS

MSP hotspots, crop exposure, revenue momentum, logistics bottlenecks, slow-for-distance routes, volume concentration, price instability, heat exposure, coverage gaps, seasonal peaks. Each returns a severity, the evidence used, and a recommended action — and each is sandboxed, so one failing check cannot break the tab.

Q&A

ASK THE DATA

Natural-language questions route to intents and are answered by computing over the dataframes, so every figure quoted is real rather than generated. 8 one-tap suggested questions cover risk, MSP, revenue, arrivals, prices, farmers, logistics and weather; many answers ship with a supporting table.

LLM

OPTIONAL CLAUDE LAYER

Paste an Anthropic API key and questions the rule-based router cannot map are escalated to the Claude API along with a compact statistical briefing of the current selection. Strictly optional — everything above works fully offline.

Why this matters in a demo: it runs with the wifi switched off, and it cannot hallucinate a statistic — it computes them.

From five broken CSVs to a targeted intervention list

TECH STACK

- Python · pandas · NumPy
- Streamlit (cached data layer)
- Plotly Express & Graph Objects
- statsmodels (OLS trendlines)
- Optional Anthropic Claude API

WHAT WE PROVED

- 40.2% of sales breach MSP
- Risk is concentrated in 3 of 57 mandis
- Delay is gate overhead, not distance
- Averages hide transaction-level harm
- 25.8% of raw data is untrustworthy

NEXT 3 SPRINTS

- Mandi-level weather sensor mapping
- SMS price alerts to farmers pre-arrival
- ARIMA forecast replacing linear trend
- Live Agmarknet API instead of CSVs
- Route optimiser on dwell-time data

The one-line pitch: MandiPulse turns unusable mandi paperwork into a ranked, weight-tunable list of the exact yards where farmers are losing money — and tells you which of price discovery, volatility or logistics to fix at each one.